

MongoDB Assignment 2

Student Record Management System — Query Outputs

The following is the terminal output produced by running the MongoDB assignment queries against the collegeDB database.

```
PS C:\Users\madda\OneDrive\Desktop\fs_lab\assignments\Assignment-2> mongosh mongodb+srv://maddalasuryaprabhath24csm_db_user:qQryVkudwQMvvoFa@cluster0.lz21ty5.mongodb.net/?appName=Cluster0 --file queries
```

```
--- 1. ALL Students ---
```

```
{
  _id: ObjectId('6ac23162c8c4101eda997f4d'),
  rollNo: '23CM001',
  name: 'Ravi Kumar',
  branch: 'CSE-AIML',
  year: 3,
  marks: 85,
  email: 'ravi@example.com'
}
{
  _id: ObjectId('6ac23172c8c4101eda997f4f'),
  rollNo: '23CM002',
  name: 'Priya Sharma',
  branch: 'CSE-AIML',
  year: 3,
  marks: 92,
  email: 'priya@example.com'
}
{
  _id: ObjectId('6ac231907266cbd62fd23418'),
  rollNo: '23CM004',
  name: 'Sneha Rao',
  branch: 'ECE',
  year: 3,
  marks: 68,
  email: 'sneha@example.com'
}
{
  _id: ObjectId('6ac231a27266cbd62fd2341a'),
  rollNo: '23CM005',
  name: 'Kiran Kumar',
  branch: 'CSE',
  year: 2,
  marks: 48,
  email: 'kiran@example.com'
}
{
  _id: ObjectId('6ac231b07266cbd62fd2341c'),
  rollNo: '23CM006',
  name: 'Ananya Reddy',
  branch: 'CSE-AIML',
  year: 2,
  marks: 88,
  email: 'ananya@example.com'
}
{
  _id: ObjectId('6ac231bd7266cbd62fd2341e'),
  rollNo: '23CM007',
  name: 'Rahul Verma',
  branch: 'ECE',
  year: 2,
  marks: 75,
  email: 'rahul@example.com'
}
```

```

    year: 2,
    marks: 55,
    email: &#x27;rahul@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231cb7266cbd62fd23420&#x27;),
  rollNo: &#x27;23CM008&#x27;,
  name: &#x27;Meghana Das&#x27;,
  branch: &#x27;CSE&#x27;,
  year: 3,
  marks: 95,
  email: &#x27;meghana@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231d77266cbd62fd23422&#x27;),
  rollNo: &#x27;23CM009&#x27;,
  name: &#x27;Vikram Singh&#x27;,
  branch: &#x27;EEE&#x27;,
  year: 3,
  marks: 72,
  email: &#x27;vikram@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231e97266cbd62fd23424&#x27;),
  rollNo: &#x27;23CM010&#x27;,
  name: &#x27;Pooja Patel&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 2,
  marks: 45,
  email: &#x27;pooja@example.com&#x27;
}

```

--- 2. CSE-AIML Students ---

```

{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;,
  name: &#x27;Ravi Kumar&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 3,
  marks: 85,
  email: &#x27;ravi@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23172c8c4101eda997f4f&#x27;),
  rollNo: &#x27;23CM002&#x27;,
  name: &#x27;Priya Sharma&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 3,
  marks: 92,
  email: &#x27;priya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231b07266cbd62fd2341c&#x27;),
  rollNo: &#x27;23CM006&#x27;,
  name: &#x27;Ananya Reddy&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 2,
  marks: 88,
  email: &#x27;ananya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231e97266cbd62fd23424&#x27;),
  rollNo: &#x27;23CM010&#x27;,
  name: &#x27;Pooja Patel&#x27;,

```

```
branch: &#x27;CSE-AIML&#x27;;
year: 2,
marks: 45,
email: &#x27;pooja@example.com&#x27;
}

--- 3. MARKS GREATER THAN 75 ---
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;;
  name: &#x27;Ravi Kumar&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 3,
  marks: 85,
  email: &#x27;ravi@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23172c8c4101eda997f4f&#x27;),
  rollNo: &#x27;23CM002&#x27;;
  name: &#x27;Priya Sharma&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 3,
  marks: 92,
  email: &#x27;priya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231b07266cbd62fd2341c&#x27;),
  rollNo: &#x27;23CM006&#x27;;
  name: &#x27;Ananya Reddy&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 2,
  marks: 88,
  email: &#x27;ananya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231cb7266cbd62fd23420&#x27;),
  rollNo: &#x27;23CM008&#x27;;
  name: &#x27;Meghana Das&#x27;;
  branch: &#x27;CSE&#x27;;
  year: 3,
  marks: 95,
  email: &#x27;meghana@example.com&#x27;
}

--- 4. SEARCH BY ROLL NUMBER ---
{
  _id: ObjectId(&#x27;6ac231a27266cbd62fd2341a&#x27;),
  rollNo: &#x27;23CM005&#x27;;
  name: &#x27;Kiran Kumar&#x27;;
  branch: &#x27;CSE&#x27;;
  year: 2,
  marks: 48,
  email: &#x27;kiran@example.com&#x27;
}

--- 5. THIRD YEAR Students ---
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;;
  name: &#x27;Ravi Kumar&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 3,
  marks: 85,
  email: &#x27;ravi@example.com&#x27;
```

```

}
{
  _id: ObjectId(&#x27;6ac23172c8c4101eda997f4f&#x27;),
  rollNo: &#x27;23CM002&#x27;,
  name: &#x27;Priya Sharma&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 3,
  marks: 92,
  email: &#x27;priya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231907266cbd62fd23418&#x27;),
  rollNo: &#x27;23CM004&#x27;,
  name: &#x27;Sneha Rao&#x27;,
  branch: &#x27;ECE&#x27;,
  year: 3,
  marks: 68,
  email: &#x27;sneha@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231cb7266cbd62fd23420&#x27;),
  rollNo: &#x27;23CM008&#x27;,
  name: &#x27;Meghana Das&#x27;,
  branch: &#x27;CSE&#x27;,
  year: 3,
  marks: 95,
  email: &#x27;meghana@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231d77266cbd62fd23422&#x27;),
  rollNo: &#x27;23CM009&#x27;,
  name: &#x27;Vikram Singh&#x27;,
  branch: &#x27;EEE&#x27;,
  year: 3,
  marks: 72,
  email: &#x27;vikram@example.com&#x27;
}

--- 6. UPDATE MARKS ---
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;,
  name: &#x27;Ravi Kumar&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 3,
  marks: 90,
  email: &#x27;ravi@example.com&#x27;
}

--- 7. UPDATE EMAIL ---
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;,
  name: &#x27;Ravi Kumar&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 3,
  marks: 90,
  email: &#x27;ravi.kumar@example.com&#x27;
}

--- 8. SORT BY MARKS DESCENDING ---
{
  _id: ObjectId(&#x27;6ac231cb7266cbd62fd23420&#x27;),
  rollNo: &#x27;23CM008&#x27;,

```

```
name: &#x27;Meghana Das&#x27;;
branch: &#x27;CSE&#x27;;
year: 3,
marks: 95,
email: &#x27;meghana@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23172c8c4101eda997f4f&#x27;),
  rollNo: &#x27;23CM002&#x27;;
  name: &#x27;Priya Sharma&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 3,
  marks: 92,
  email: &#x27;priya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;;
  name: &#x27;Ravi Kumar&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 3,
  marks: 90,
  email: &#x27;ravi.kumar@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231b07266cbd62fd2341c&#x27;),
  rollNo: &#x27;23CM006&#x27;;
  name: &#x27;Ananya Reddy&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 2,
  marks: 88,
  email: &#x27;ananya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231d77266cbd62fd23422&#x27;),
  rollNo: &#x27;23CM009&#x27;;
  name: &#x27;Vikram Singh&#x27;;
  branch: &#x27;EEE&#x27;;
  year: 3,
  marks: 72,
  email: &#x27;vikram@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231907266cbd62fd23418&#x27;),
  rollNo: &#x27;23CM004&#x27;;
  name: &#x27;Sneha Rao&#x27;;
  branch: &#x27;ECE&#x27;;
  year: 3,
  marks: 68,
  email: &#x27;sneha@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231bd7266cbd62fd2341e&#x27;),
  rollNo: &#x27;23CM007&#x27;;
  name: &#x27;Rahul Verma&#x27;;
  branch: &#x27;ECE&#x27;;
  year: 2,
  marks: 55,
  email: &#x27;rahul@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231a27266cbd62fd2341a&#x27;),
  rollNo: &#x27;23CM005&#x27;;
  name: &#x27;Kiran Kumar&#x27;;
```

```

branch: &#x27;CSE&#x27;;
year: 2,
marks: 48,
email: &#x27;kiran@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231e97266cbd62fd23424&#x27;),
  rollNo: &#x27;23CM010&#x27;;
  name: &#x27;Pooja Patel&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 2,
  marks: 45,
  email: &#x27;pooja@example.com&#x27;
}

```

```

--- 9. CREATE INDEX ON rollNo ---
rollNo_1

```

```

--- 10. DISPLAY INDEXES ---
[
  {
    v: 2,
    key: {
      _id: 1
    },
    name: &#x27;_id_&#x27;
  },
  {
    v: 2,
    key: {
      rollNo: 1
    },
    name: &#x27;rollNo_1&#x27;
  }
]

```

```

--- 11. INDEX SEARCH DEMONSTRATION ---
{
  explainVersion: &#x27;1&#x27;;
  queryPlanner: {
    namespace: &#x27;collegeDB.Students&#x27;;
    parsedQuery: {
      rollNo: {
        &#x27;$eq&#x27;; &#x27;23CM005&#x27;
      }
    },
    indexFilterSet: false,
    queryHash: &#x27;B62BB36F&#x27;;
    planCacheShapeHash: &#x27;B62BB36F&#x27;;
    planCacheKey: &#x27;E3E3A3E6&#x27;;
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: &#x27;FETCH&#x27;;
      inputStage: {
        stage: &#x27;IXSCAN&#x27;;
        keyPattern: {
          rollNo: 1
        },
        indexName: &#x27;rollNo_1&#x27;;
      }
    }
  }
}

```

```
    isMultiKey: false,
    multiKeyPaths: {
      rollNo: []
    },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: &#x27;forward&#x27;;
    indexBounds: {
      rollNo: [
        &#x27;[&quot;23CM005&quot;;, &quot;23CM005&quot;]&#x27;
      ]
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 1,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: &#x27;FETCH&#x27;;
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 1,
    alreadyHasObj: 0,
    inputStage: {
      stage: &#x27;IXSCAN&#x27;;
      nReturned: 1,
      executionTimeMillisEstimate: 0,
      works: 2,
      advanced: 1,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      keyPattern: {
        rollNo: 1
      },
      indexName: &#x27;rollNo_1&#x27;;
      isMultiKey: false,
      multiKeyPaths: {
        rollNo: []
      },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: &#x27;forward&#x27;;
      indexBounds: {
        rollNo: [
```

```

        &#x27;[&quot;23CM005&quot;;, &quot;23CM005&quot;]&#x27;
    ]
},
keysExamined: 1,
seeks: 1,
dupsTested: 0,
dupsDropped: 0
}
}
},
queryShapeHash: &#x27;4CCF7490C115A30D0FB7B6E06FAAFE4D743EF024771290F7E06C64DCD67AD61E&#x27;;
command: {
  find: &#x27;Students&#x27;;,
  filter: {
    rollNo: &#x27;23CM005&#x27;
  },
  &#x27;$db&#x27;;: &#x27;collegeDB&#x27;
},
serverInfo: {
  host: &#x27;ac-vwn4ddn-shard-00-01.lz21ty5.mongodb.net&#x27;;,
  port: 27017,
  version: &#x27;8.0.34&#x27;;,
  gitVersion: &#x27;51db4515a6597aedcf6188ee4e48a27e65b065ae&#x27;
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: &#x27;trySbeRestricted&#x27;;,
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
&#x27;$clusterTime&#x27;;: {
  clusterTime: Timestamp({ t: 1791116407, i: 18 }),
  signature: {
    hash: Binary.createFromBase64(&#x27;d/39wY6Q9GeuUc0vZq8Bso7gURo=&#x27;;, 0),
    keyId: Long(&#x27;7646876545369243657&#x27;)
  }
},
operationTime: Timestamp({ t: 1791116407, i: 18 })
}

--- 12. ABOVE 80 ---
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;)),
  rollNo: &#x27;23CM001&#x27;;,
  name: &#x27;Ravi Kumar&#x27;;,
  branch: &#x27;CSE-AI&#x27;;,
  year: 3,
  marks: 90,
  email: &#x27;ravi.kumar@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23172c8c4101eda997f4f&#x27;)),
  rollNo: &#x27;23CM002&#x27;;,
  name: &#x27;Priya Sharma&#x27;;,
  branch: &#x27;CSE-AI&#x27;;,
  year: 3,
  marks: 92,

```

```
    email: &#x27;priya@example.com&#x27;
  }
  {
    _id: ObjectId(&#x27;6ac231b07266cbd62fd2341c&#x27;),
    rollNo: &#x27;23CM006&#x27;,
    name: &#x27;Ananya Reddy&#x27;,
    branch: &#x27;CSE-AIML&#x27;,
    year: 2,
    marks: 88,
    email: &#x27;ananya@example.com&#x27;
  }
  {
    _id: ObjectId(&#x27;6ac231cb7266cbd62fd23420&#x27;),
    rollNo: &#x27;23CM008&#x27;,
    name: &#x27;Meghana Das&#x27;,
    branch: &#x27;CSE&#x27;,
    year: 3,
    marks: 95,
    email: &#x27;meghana@example.com&#x27;
  }
}
```

--- 13. BELOW 50 ---

```
{
  _id: ObjectId(&#x27;6ac231a27266cbd62fd2341a&#x27;),
  rollNo: &#x27;23CM005&#x27;,
  name: &#x27;Kiran Kumar&#x27;,
  branch: &#x27;CSE&#x27;,
  year: 2,
  marks: 48,
  email: &#x27;kiran@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231e97266cbd62fd23424&#x27;),
  rollNo: &#x27;23CM010&#x27;,
  name: &#x27;Pooja Patel&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 2,
  marks: 45,
  email: &#x27;pooja@example.com&#x27;
}
}
```

--- 14. HIGHEST SCORING STUDENT ---

```
{
  _id: ObjectId(&#x27;6ac231cb7266cbd62fd23420&#x27;),
  rollNo: &#x27;23CM008&#x27;,
  name: &#x27;Meghana Das&#x27;,
  branch: &#x27;CSE&#x27;,
  year: 3,
  marks: 95,
  email: &#x27;meghana@example.com&#x27;
}
}
```

--- 15. CSE-AIML Students ---

```
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;,
  name: &#x27;Ravi Kumar&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 3,
  marks: 90,
  email: &#x27;ravi.kumar@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23172c8c4101eda997f4f&#x27;),
}
```

```
rollNo: &#x27;23CM002&#x27;;
name: &#x27;Priya Sharma&#x27;;
branch: &#x27;CSE-AIML&#x27;;
year: 3,
marks: 92,
email: &#x27;priya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231b07266cbd62fd2341c&#x27;),
  rollNo: &#x27;23CM006&#x27;;
  name: &#x27;Ananya Reddy&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 2,
  marks: 88,
  email: &#x27;ananya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231e97266cbd62fd23424&#x27;),
  rollNo: &#x27;23CM010&#x27;;
  name: &#x27;Pooja Patel&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 2,
  marks: 45,
  email: &#x27;pooja@example.com&#x27;
}

--- 16. Students SORTED BY MARKS ---
{
  _id: ObjectId(&#x27;6ac231cb7266cbd62fd23420&#x27;),
  rollNo: &#x27;23CM008&#x27;;
  name: &#x27;Meghana Das&#x27;;
  branch: &#x27;CSE&#x27;;
  year: 3,
  marks: 95,
  email: &#x27;meghana@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23172c8c4101eda997f4f&#x27;),
  rollNo: &#x27;23CM002&#x27;;
  name: &#x27;Priya Sharma&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 3,
  marks: 92,
  email: &#x27;priya@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac23162c8c4101eda997f4d&#x27;),
  rollNo: &#x27;23CM001&#x27;;
  name: &#x27;Ravi Kumar&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 3,
  marks: 90,
  email: &#x27;ravi.kumar@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231b07266cbd62fd2341c&#x27;),
  rollNo: &#x27;23CM006&#x27;;
  name: &#x27;Ananya Reddy&#x27;;
  branch: &#x27;CSE-AIML&#x27;;
  year: 2,
  marks: 88,
  email: &#x27;ananya@example.com&#x27;
}
{
```

```
_id: ObjectId(&#x27;6ac231d77266cbd62fd23422&#x27;),
rollNo: &#x27;23CM009&#x27;,
name: &#x27;Vikram Singh&#x27;,
branch: &#x27;EEE&#x27;,
year: 3,
marks: 72,
email: &#x27;vikram@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231907266cbd62fd23418&#x27;),
  rollNo: &#x27;23CM004&#x27;,
  name: &#x27;Sneha Rao&#x27;,
  branch: &#x27;ECE&#x27;,
  year: 3,
  marks: 68,
  email: &#x27;sneha@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231bd7266cbd62fd2341e&#x27;),
  rollNo: &#x27;23CM007&#x27;,
  name: &#x27;Rahul Verma&#x27;,
  branch: &#x27;ECE&#x27;,
  year: 2,
  marks: 55,
  email: &#x27;rahul@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231a27266cbd62fd2341a&#x27;),
  rollNo: &#x27;23CM005&#x27;,
  name: &#x27;Kiran Kumar&#x27;,
  branch: &#x27;CSE&#x27;,
  year: 2,
  marks: 48,
  email: &#x27;kiran@example.com&#x27;
}
{
  _id: ObjectId(&#x27;6ac231e97266cbd62fd23424&#x27;),
  rollNo: &#x27;23CM010&#x27;,
  name: &#x27;Pooja Patel&#x27;,
  branch: &#x27;CSE-AIML&#x27;,
  year: 2,
  marks: 45,
  email: &#x27;pooja@example.com&#x27;
}
```